



SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI

Mechanical

1A/1B/1C Schedule of 3Phase Electric Locomotive [Inspection]

Loco no. 279-23

लोको संख्या Loco No. 32456 क्लास Class 1C शिड्यूल लेने का दिनांक Date taken under Sch 1VS

	आगमन Incoming / शिड्यूल Schedule		अंतिम निरीक्षण Final Inspection	
	अपर डेक Upper Deck	अंडर ट्रक Under Truck	अपर डेक Upper Deck	अंडर ट्रक Under Truck
Date/ Shift	23.9.24 11:00 AM	26.9.24 08:00 AM		
Name of Technician	monu	m. dufin		
Designation/ Token No.	21m-IST	MLC		
Signature				

Customer Feed Back/Bookings (Record feedback given by Loco pilot)

Sr. No.	Customer Feed Back/Bookings	Action Taken	Name of TCN / Attended By	Remarks, if any & Sign. of JE/SSE
1. Panto - 1 Lowering time adjustment.				
2. Cab-1 wiper blade broken. both LP & ALP				
3. Cab-1 B.P hose Pipe cut mark.				
4. Cab-1 B.P pipe hold clamp broken.				
5. CPA make pressure taken access time.				
6. Send box dummy missing wheel no - 02108				
7. Both TF + silencer Pinic.				
8. Cab-1 CP wiper water jet n/w.				
9. All send not working				
10. Cab-1 F.P hose Pipe leakage from clec nut.				
11. Cab-2 RIS valve handle to adjust				
12. Cab-2 ALP wiper adjust				
13. Cab-2 ALP wiper blade broken				
14. Cab-2 Both wiper jet not working.				
15. Cab-2 Both wiper jet not working.				
16. Wheel no. send pipe misalignment				
17. DS open करने पर न्यू मेकनिकल पैनल से आवाज आता है और DS बंद करने पर बज नहीं आता है				
18.				

Inspection Wing Mechanical: Check and record Following

Std. Value	CPA Loading Time (Pressure 0 to 8.0 kg/cm ²)	Safety Valve Blowing	Pantograph-1 Raising	Pantograph-1 Lowering	Pantograph-2 Raising	Pantograph-2 Lowering	Compressor Loading Time (Pressure 8 to 9.5 kg/cm ²)	CP Safety Valve Blowing	Unloaded valve
			Second	15 Second	Second	15 Second	Comp-1	Comp-2	Operation time Approx. 12 Sec.
GI	8.6	7.2	25	6.5	6.4	50 sec	59 sec	7.11	04 sec
Final	8.7	7.4	11.2	6.5	6.5	52 SEC	57 SEC	7.11	04 SEC

Std. Value	MR safety Valve pressure		Duplex check valve setting/Brake Pipe Pressure by Auto Brake (AB)		Feed Pipe Pressure		Brake Cylinder Pressure by Direct Brake (SAB)		Horn Sound			
	Blowing	Closing	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	LP	ALP	LP	ALP
GI	10.5	9.5	5.0	5.0	5.0	5.0	3.5	3.5	OK	OK	OK	OK
Final	11	9.5	5.0	5.0	5.0	5.0	3.5	3.5	OK	OK	OK	OK

Std. Val.	Computerized Control Brake (CCB) System if applicable		Air Flow	Air Dryer		ADC	MRC Time By both CP 0-10kg/cm ²	
	Alert and Penalty	Working	Working	Working, Tower to change every minute (FTL, SIL) & in every two minutes (KSL)	Humidity Indicator Colour Std-Blue	Working	3.5 Minutes (Max.) For 1750 LPM or [8.5 Minute for each 1450 LPM CP]	
GI	Passed successfully	Working	OK	Working	blue	Working	02 min 10 sec	
Final				Bag Polymer			03.20	

18. MR setting रही लोको से 10.5 पर कट आ रहा है

19. Cab-2 gauge light काटि नहीं कर रहा है

20. CP-2 oil level oil नीचा नहीं है

21. Air dryer isolate हो चुका है

Signature of JE/SSE

22. रिम 44 से बोझ उतार का निषेध है

	CPA Pressure build up time 8.5 kg/cm ²	BP drop in 5 Minute	FP drop in 5 Minute	MR drop in 15 Minute	Press BPPB to apply or release parking brake. Check Pressure in Parking Brake gauge	VCB Pressure switch setting
Std. Value	60 Sec. Max.	Max. drop 0.15 kg/cm ²	Max. drop 0.70 kg/cm ²	Max. drop 1.0 kg/cm ²	6.0 ± 0.15 kg/cm ²	Opens 4.5 ± 0.15 kg/cm ² , Closes 5.5 ± 0.15 kg/cm ²
GI	01 min 13 sec	No drop	No drop	0.6		
Final	01.17	No drop	No drop	0.7		

	ACP on 4mm test plate	Auto brake controller Position	BP (kg/cm ²)	BC (kg/cm ²)
			Cab-1	Cab-2
Std Values	Working	Run	5.0 ± 0.10	5.0
		Initial Application	4.6 ± 0.10	4.7
	Cab-1 Cab-2	Full Service	3.35 ± 0.20	3.4
		Emergency	Below 0.30	0.0
Gen. Insp.	work	working		
Final	u	u	ok	ok

	Loco pilot cabin condition	Wiper Operation	Look out Glass	Fire Extinguisher	Sand box	Sander Operation	Cleaning of Loco
	Cab-1 Cab-2	Cab-1 Cab-2	Cab-1 Cab-2			Cab-1 Cab-2	
Std. Values	Good Good	Working Working	Clear Clear	Sealed	Full	Working	Cleaned
Gen. Insp.	dirty dirty	OK OK	n/c n/c	sealed	empty	empty	n/c
Final	ok ok	ok ok	ok ok	ready	full	full	ok

क्र.स.	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल)	कार्यवाही की गयी	टीसीएन का नाम	हस्ताक्षर/ टिप्पणी
SN	Detail of Work/ Inspection (Schedule)	Action Taken	Name of TCN	Sign/ Remarks
(A) Pneumatic System (When Loco is energised check, attend and record)				
1	Check correctness of reading of BP, FP, MR, BC, AFI & RS gauges.	check		
2	Check on dummy of BP/FP pipe.	check		
3	Check the working of PTDC			
4	Check the proper functioning of all the gauges of the driver desk BP: (i) Normal (ii) with 7.5 mm hole BP should not drop below 4 kg/cm ² in 60 second. FP: (i) Normal (ii) with 5.5 mm hole FP should not drop below 4.5 kg/cm ² in 60 second. MR: (i) Mainline MR leakage without application of SA-9 Direct Brake (ii) Mainline MR leakage with application of SA-9 Direct Brake	BP = 4.7 F.P = 6.0		ment
5	(RS drop) Panto line air leakage 0.70 kg/cm ² in 5 minutes.	check		
6	Check working of Panto selection switch.	check		
7	Position of isolating cock on pneumatic panel. (70, 74, 136 in open condition & 47 in close condition.)	check		
8	Drain moisture from MR and Inter Cooler.	check		
9	Check functioning of air dryer for purging, cycle time and change over with MR pressure of 8.0 kg/cm ² .	check		
10	Check Position of the Air dryer cock (13, 14 in service & 187 in isolated)	check		
11	Check air dryer charging when compressor is in working. (FTIL -120 sec/SIL-180 sec)	check		
12	Check and attend leakage of air dryer, compressor valves, Pneumatic valves, Pneumatic panels & pipelines	check		

Signature of QAG JE/SSE

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण /(शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
B	BRAKE CONTROLLER (AUTO AND DIRECT)			
1	Check the functioning of driver's direct air brake & automatic train brake.	ch-1 on		
2	Check the free movement of the brake controller.	ch-1 on		
3	Perform repeated application and release of brakes with both auto and direct brake controller from both cabs and ensure that BP & BC pressure build up & drop is proper on the gauge. Ensure by physically going to individual main and parking brake cylinder that their operation is smooth and without air leakage.	ch-1 on		
C	CARBODY FILTER:			
	Check condition and clean car body filter and refit.	cleaned		
1	Thoroughly clean all roof and side body filters with water jet.	cleaned		
2	Check tightness of Roof joint clamp and Roof top bolt.	checked		
3	Remove Louvers & Fly screen and clean it. Repair if found defective.	checked		
	IC sch.			
4	Remove the tread plate and clean the drainage channels. Check and adjust, if necessary, the door-operating rod.	clean		
H	LOOKOUT GLASS & SIDE WINDOWS			
1	Clean any built up dirt and debris from the channel below the cab-sliding window. Clear the sliding window channel drains hole in the door opening.	—		
2	Clean look out glass with detergent/water solution.	—		
I	WASHER / WIPER			
1	Check the operation of windscreen wipers using the manual handles. Rectify any fault found.	ch-1 on		
2	Check the condition of windscreen wiper blades. Replace if required.	ch-1 on		
4	Check the water level in the reservoir at No.1 and No.2 end. Top up if required. Check wipers for proper operation.	ch-1 on		
6	Check for damage of air and water pipes.	ch-1 on		
7	Examine all components for wear, looseness scoring and cracks, renew any defective components.	ch-1 on		
8	Check seat-mounting points for integrity and ensure fixing or fully tightened.	ch-1 on		
J	Key interlocking: Check all keys are present & the complete system is operational.	ch-1 on		
	(IC Sch)			
9	Check the wiper and parallel arm assembly for wear and damage. Replace the arm or any part if there is excessive wear or damage.	ch-1 on		
10	Clean the washer nozzle jet. (IC Sch)	—		
K	Fire Extinguisher			
1	Ensure availability of fire extinguishers and ensure that it is not expired.	ch-1 on		
2	Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary.	ch-1 on		
3	Clean pipeline by compressed air and check that there should not be any leakage.	—		
4	CO2 Gas pressure and date: Cab-1 30/12/24 80 Psi/cm ² Cab-2 30/12/24 82 Psi/cm ²	—		
5	Clean fire detection unit pipeline air intake holes with soft non-metallic brush & vacuum cleaner.	—		

30.12.24

Signature of QAC JE/SSE

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
L	OIL LEAKAGE IN MACHINE ROOM / CLEANING OF MACHINE ROOM.			
1	Visually inspect the machine room for any signs oil leakage from traction converter, transformer, D.C. Link, and series resonant circuit capacitors and other electrical equipment and carefully clean the same. (Caution : Electric supply room)	check on		
2	Clean the complete machine room with the air pressurized supply and vacuum cleaner to remove all the dust particles.			
M	SAND GEAR SYSTEM			
	a) Check Sand box, cover gasket, Ejector and replace gasket if necessary.	check on		
	b) Check all sander pipe for alignment and check fasteners also.	check on		
	c) Check and ensure tightness of Sand box foundation bolt.			
	d) Ensure sander pipe height 60mm above rail level.			
	e) Refill sand in all sand boxes.			
N	OIL COOLING UNIT CASING WITH RADIATOR			
1	Remove all dust, dirt and debris from the radiator chamber via the machine room access cover. (By using vacuum cleaner)	check		
2	Visually check the conservator tank along with its gauge glass for any sign of crack / damage / oil leakage.	check		
3	Measure air delivery before and after cleaning of radiator.	check		
4	Measure vibrations of oil cooling casing using vibration analyzer and take appropriate action.	check		
5	Visually check the casing and support arrangement for oil pumps on the casing for any sign of crack or the damage.	check		
6	Visually check the oil cooler radiator for any oil leakage from VPTFP and VPSR and external damage from top to bottom.	check		
	IC Sch			
7	a) Clean radiator with hot water jet and check that the radiator is not blocked.	check		
8	b) Tighten the front foundation bolts of casing mounted on to the radiator and check intactness of rail and Z-clamp welding crack.	check		

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UPPER DECK [Inspect, Attend and Record]			
SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts.	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Condition and working of all pressure gauges.	✓	
4.	Working of all brake valves.	✓	
5.	Leakage of air (MR)	✓	
6.	Gauges, securing of loco pilot cabin.	✓	
7.	Operation of Wipers.	✓	
8.	Co-Action working of A9 Auto brake.	✓	
9.	Fire Extinguishers (prescribed Nos.)	✓	
10.	Check tightness of compressor foundation bolt & abnormal sound	✓	02 + 02 Big small
11.	check slackness of compressor fan and no vibration in comp.	✓	
12.	Check tightness of air dryer foundation bolt.	✓	
13.	Ensure Air dryer indicator blue colour	✓	
14.	Ensure clamping of all pipe to avoid vibration and rubbing.	✓	
15.	Ensure fitment of all split pins	✓	
16.	Check tightness of all bolts of Loco Roof	✓	
17.	Check window of both cab, shutter window door & corridor doors	✓	

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Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts	✓	
2.	Clamping and condition of its welding/fastener	✓	
3.	Air leakage in bogie	✓	
4.	Overheated parts / Abnormal wear in parts	✓	
5.	Check and Ensure Brake block and Wheel clearance >10mm	✓	

(Remarks if any & sign):

Signature and Name of Technician for Inspection Incoming with date & shift:

Mantu + Rakesh
Mantu 23.9.24

Signature and Name of JE/SSE for Inspection Incoming with date & shift:

Signature and Name of Technician for Inspection Final with date & shift:

Signature and Name of JE/SSE for Inspection Final with date & shift:

अन्य किसी भी जानकारी Any Other Information:

Signature of QAG SSE Incharge (Inspection)

Signature of QAG JE/SSE

SCHEDULE PROGRESS CHART

SN	Date	Shift	QC	Work Details	Name of JE/SSE	Sign of JE/SSE	Remarks
1	23.9.24	16/24		Pre inspection			
2	24/9/24	LT 8/17		com sel & Transformer done.			
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							

Signature of QAG JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3 PHASE ELECTRIC LOCOMOTIVE [UNDERFRAME]

लोको संख्या Loco No	32256	क्लास Class	IC	शिड्यूल में लेने का दिनांक Date taken under Sch.	24.9.24
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UNDERFRAME IA/IB/IC SCHEDULE & RECORDS

A-Check and record Hoot Wear, Flange Wear & Tread Wear (Record Wheel Diameter in Schedule or After TT)									
Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1	10667	2	1	2	7	10665	25	1	2
2	10669	35	15	15	8	10680	15	.5	1.5
3	10677	25	.5	3	9	10681	25	.5	2
4	10681	25	.5	1	10	10684	1	.5	1
5	10685	25	.5	2.5	11	10691	3	1	2
6	10681	3	1	1.5	12	10693	3	1	2

B- Check and Record Vertical & lateral clearance (Primary & Secondary) as per TC/0082 (REV-1) dated 17/09/2015

Standard Value	Primary vertical clearance (Cab-1 & 2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
27 to 35 mm	3200	3110	3270	3430	3490	3520	3310	3590	3440	3430	3220	3500

Standard Value	Primary lateral clearance (Cab-1 & 2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
15 to 22 mm	1750	1600	1590	1720	1650	1550	1530	1650	1610	1570	1750	1610

Standard Value	Secondary Vertical clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-2 ALP Side
32 to 60 mm	3710	3650	3820	3650

Standard Value	Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-2 ALP Side
45 to 55 mm	4530	4735	44790	4810

C- Check and record wheel gauge below Parameters: (IC Schedule) WHEEL GAUGE [Permissible limit 1596+3-0.5 mm, (1596±0.5 mm for RB/New loco)] checked by V/S Section

Axle No.	A	B	C	Average	Remarks
1					
2					
3					
4					
5					
6					

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE

Check and record below Parameters in Schedule or if TT is done.

Axle No.	Standard (mm)		Cab-1		Cab-2	
	Max	Min	R1	L1	R2	L2
Location						
Buffer Length	635	615	642	640	642	638
Buffer Height	1105	1030	1080	1065	1085	1090
Rail guard height	118	115	114	120(114)	111	105
CBC Height	1105	1030	1062	1065	1073	

		Cab-1	Cab-2
STRIKER WEAR PLATE	New 8 mm, Condemn 2 mm		
SHANK WEAR PLATE	New 6 mm, Condemn 1 mm		

E- Measure D SHAKLE and write Coupling No.

CAB-1

CAB-2

Check and Record Gear case Oil Level:

TM	OIL FOUND	OIL TOP UP	TM	OIL FOUND	OIL TOP UP
1	Full	Full	4	Full	Full
2	5	1 Lit	5	5	1 Lit
3	Full	Full	6	Full	Full

	Hand Brake Operation	CB Coupler Operation & Condition		Transition Screw Coupling Operation & Condition		CBC Lock Pin		HP/FF Pipe	
Std. Value	Working	Normal		Normal		Intact		Intact	
		Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2
Gen. Insp.		OK	OK	OK	OK	OK	OK	OK	
Final Insp.		OK	OK	OK	OK	OK	OK		

Brake Piston Travel :						Brake Shoe release	Standard 3/9/10 mm
Class of Loco	Standard value	Observed (mm)					
WAG7/WAP7	107 to 117 mm	Location-	1	2	3	4	
WAG9		Location-	115	114	110	112	

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
F	DRAFT GEAR AND COUPLERS		
	1. Liberally apply general-purpose grease to the face of the center buffer coupler at the end of the Locomotive	Cham	
	2. Check the intactness of the CBC bottom support plate bolts	Cham	
	3. Check intactness and functioning of CBC operating handle.	Cham	
	4. Check visually the striker casting of CBC assembly.	Cham	
	5. Check for working of CBC, Lock and unlock it, locking arrangement, working of TC	Cham	
	6. Check washer lock at knuckle pin bottom	Cham	
	7. Check gap between Striker and Draft gear for draft gear slackness	Cham	
	IC Sch.		
	8. Check the knuckles against crack by DPT	Cham	
	9. Check knuckle by gauge no. 1&2 (130-135 mm by go no go gauge)	Cham	
	10. Check knuckle nose wear by gauge no. 3 go no go gauge	Cham	
G	BUFFERS		
	1. Check buffers for any crack visually of the flange and intactness of foundation bolts, check nuts and split pins (4 nos. each).	Cham	

Signature and Name of Technician

क्यूजी जेई/एसएसई QAG-JE/SSE

→ grease no. (1) to (5) buffer valve cleaning done! 23/09

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	2. Liberally apply general purpose grease to the face of the buffers at the ends of the Locomotive. Also smear grease on the buffer head plunger.		
	IC Sch.		
	3. Conduct RDPT of Buffers to detect crack flaws.		
	4. Check and measure the buffer height and buffer length.		
H	SUSPENSION		
	1. Examine the primary and secondary springs for signs of damage, cracks and broken end visually with torch light and tapping by a hammer & replace, if necessary		
	2. Examine different types of dampers for oil leakage and intactness of fasteners and end fittings		
	3. Check the condition of various Spheris block in the primary and secondary suspension arrangement and of various equipment mounted in the under frame like axle guide link, traction motor and its support arm, gear case support arm for their healthiness (it should not be crack) and replace, if necessary		
	IC Sch.		
	4. Proper torque of all fasteners related to suspension arrangement and mounting of equipment in under frame		
	5. RDPT of all TM nose and its support arm		
I	TRACTION BAR		
	1. Check the well being and presence of the safety sling		
	2. Check the push/pull rod pivot heads for damage		
	3. Examine the traction link assembly for signs of damage. Any defective item should be renewed		
	4. Check the fastener at the traction ends and those retaining the traction rod for security. If a fastener is loose, it must be renewed and tightened to the specified torque. Ensure intactness of the locking plate. SMI 241		
	5. RDPT of traction bar at end (flange) portion & V Ring		
	6. Housing flange		
	7. Examine limit chain and links for wear, corrosion or damage		
	8. Inspect the safety slings, pins and clips for wear and damage		
	9. Check Aclathan/Vulkollon ring against any crack or shifting		
	10. Check gap in housing flange and traction link flange		
	11. Torque modified bolt M20 X65 by 260Nm (SMI 241)		
	IC Sch.		
	12. RDPT of Pivot post for any crack		
J	BOGIE FRAME AND BRAKE GEAR		
	BRAKE BLOCK		
	1. Inspect the brake blocks for wear. If it is below the minimum limit, replace the complete locomotive set brake block.		
	2. Check the brake block key security		
	3. Check the brake rigging for loose, damaged or worn out parts Rectify as required.		
	4. Examine the bogie frame, transom beam & damper base for signs of crack/damage		
	5. Check fitment of Palm Pull rod J Bracket, hanger lever and TM safety rope wire.		
	6. Inspect the rubber bumps for damage deterioration. Replace if required		
	7. Examine all bogies mounted equipment (Including earthing straps) for signs of damage and security of fastening		
	8. Inspect the brake rigging rotation points and component contact surfaces. Rectify any faults found.		
	9. Check the earthing shunts provided on bogie and body.		
	Check intactness of axle box cover bolts and sealing.		

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	10. Inspect brake gear for any worn out loose, damaged, hanging or dragging parts. Confirm intactness	Clear	24/9/25
	11. Test for correct operation of the brake cylinder actuator unit.	Clear	
	12. Check the proper tightness of parking brake & service brake pneumatic pipes.	Clear	
	13. Check the external hitting marks & physical damage on the brake cylinder actuator unit.	Clear	
	14. Check the condition of parking brake cylinder bellows If damaged, replace them.	Clear	
	15. On parking brake cylinder pulling spindle & ring should be intact.	Clear	
	16. Check proper functioning of parking brake & service brake cylinder.	Clear	
	17. Check the operation (manual releasing) of the parking brakes	Clear	
	18. Check the bogie isolating cock, should be in proper position.	Clear	
	19. Check the wear of scrubber blocks, replace if required (WAP/Only)	Clear	
	20. Maintenance of Traction Motor support plate and bogie nose to Prevent crack/breakage of Traction Motor support plate (Holder for Traction motor suspension) should be done as per RDSO's SM no. RDSO/2011/EL/SM/0269 (Rev. '0') dtd, 18.05.11	Clear	
	IC Sch.	Clear	24/9/25
	21. Inspect the bogie body safety pin eye-hole at the bogie frame for wear or cracks	Clear	
	22. Maintain the vertical & lateral clearances (Primary & secondary) as per TC-082	Clear	
ii	WHEEL SET AND AXLE BOX	Clear	
	1. Examine all the wheels visually for following defects.	Clear	
	a) Cracks: if any, requires replacement.	Clear	
	b) Flats: Length 40mm or more requires re-profiling.	Clear	
	c) Cavities: Isolated cavities of length 15mm or more or multiple cavities of length 10mm or more and less than 50mm apart require re-profile/replacement.	Clear	
	d) Metal builds up Height 1mm, or above and length 50mm or more requires re-profiling. For length less than 50mm, remove the build up with hand tools.	Clear	
	e) Scoring and Grooving: Circumferentially around the wheel tread due to action of brake blocks require re-profile/replacement.	Clear	
	2. Grease Axle box after 300000 KM Engine Run	Clear	
	3. Check the temperature of axle boxes. The axle boxes which are running warm or reported to have higher temperature in comparison with others shall be attended on priority by physically opening of axle box cover and outer angle ring. The suitable maintenance shall be done either in-situ or by dismantling the axle box.	Clear	
	IC Sch.	Clear	24/9/25
	4. Measure and record the wheel diameters and tread profile. Refer RDSO's Tech. Circular no. ELRS/TC/0083 (Rev. "0"),	Clear	
	5. Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or deterioration.	Clear	
	6. Check the axle for any cracks with the help of ultrasonic flaw detector. At the same time visual inspection should be done. (IC Sch Every 06 month)	Clear	
iii	GEAR CASE (WAP7/WAG9/WAG9H)	Clear	
	1. Externally clean the gear case & its gauge glass. Inspect for any signs of oil leakage and any external hits/ damage. Verify integrity of gauge glass protective cover and clean oil sight glass for visibility.	Clear	
	2. Check the oil level in the gauge glass and top up if required. Ensure intactness of the drain plug along with its sealing. Check its tightness.	Clear	
	3. Check for any breakage of bolt and replace	Clear	
	4. Check for intactness of all fasteners of gear case	Clear	
	5. Tighten all gear case fasteners at prescribed torque	Clear	

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
6.	Check the gear case oil visually and replace if found dirty or dust full level in the gear case and fill to the requirement	Chauhan	
7.	Check Gear case oil for metal contamination	Chauhan	
8.	Visually examine for fractures, the tyre profile for wear, flat, and skidding marks	Chauhan	
9.	Check wheel root, flange and Tread wear & record Change all unserviceable Brake Blocks	Chauhan	
10.	Check condition and operation of Hand Brakes. Lubricate & adjust as necessary.	Chauhan	
11.	Adjust Brakes and lubricate union	Chauhan	
12.	Lubricate Brake Gearing, Slack Adjusters, Buffers, CB Coupler Assembly	Chauhan	
13.	Split the cotter pin, if new provided	Chauhan	
IC Sch.		Chauhan	
14.	a) Axle box old grease to remove and New grease to fill, old grease sample to be sent Lab for metal content check	Chauhan	
	b) Lubricate suspension roller tube bearing.	Chauhan	
	c) Check Transition screw coupling long and short shackle bar thickness by Go-No Go gauge (39.5 mm snap 1/25 mm snap 8)	Chauhan	
	d) Carry out DFT of Buffer Foundation and DPT/MPT of CBC Knuckle clevis & Yoke. Check size of Clevis, Knuckle, coupling by Go-No Go Gauge as specified. As per SMI-213/MIG-76	Chauhan	

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action takes in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB No.	Remarks
1.	Measure and record buffer & rail guard height.	✓	
2.	If skidded, fill Wheel Sledding Complain Report Form, FRMB/22 or if Biased wheel wear, fill up Check last for Based Wheel Wear Cause invest, FILM/24)	✓	
3.	Proper Fitment of Brake Block and as Key	✓	
4.	Rubbing of brake block with wheel in brake release position	✓	
5.	Movement of brake pull rod in brake applied position	✓	
6.	Brake Shoe and Brake hanger Pin/Bolts	✓	
7.	Brake piston movement.	✓	
8.	Support arrangement for brake cross tie bars	✓	
9.	CBC Drooping.	✓	
10.	Operation of both Transition screw couplings	✓	
11.	Operation of CBC, Operating Rod and Coupler lock Pin	✓	
12.	Operation of CBC Knuckles	✓	
13.	CBC retaining plate and support plate bolts	✓	
14.	Side buffers and their mounting bolts.	✓	
15.	Bolts of Cattle guards and back supports.	✓	
16.	Fitment of Rail Guard	✓	
17.	Gear case mounting bolts and 'C' clamp bolts	✓	
18.	Suspension bearing cap bolts and it's sealing.	✓	
19.	Crack in TM Suspension taper roller bearing housing	✓	

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE

After TT Reading

Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1					7				
2					8				
3					9				
4					10				
5					11				
6					12				

Axle No.	Cab-1		Cab-2	
Location	RI	LI	R2	L2
Buffer Length				
Buffer Height				
Rail guard height				
CBC Height				

Signature and Name of Technician

क्यूएजी जेई/एसएसई QAG-JE/SSE



पश्चिम रेलवे
Western Railway

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3PHASE ELECTRIC LOCOMOTIVE
[COMPRESSOR & PNEUMATIC SYSTEM]

लोको संख्या Loco No 32456 क्लास Class IC शिड्यूल में लेने का दिनांक Date taken under Sch 24.9.24

क्र.सं. SN	कार्यवाही कार्य / विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
A	मेन कंप्रेसर और बेबी कंप्रेसर Main Compressor and Baby Compressor			
1	Check the pressure build up timing of each compressor independently in pre-inspection and take needful action during inspection. [From 0 to 10 kg/cm ² i) With 1750LPM Comp-7 Minutes max. ii) With 1450LPM Comp- 8.5 Minutes max.]	CP-I - 06 Min. CP-II - 06 min 55 Sec		}
2	Feel the vibrations of the compressor in pre-inspection and taken needful action during inspection.	check ok		
3	Check for any external damage / abnormal sound / proper functioning of the compressor.	check ok		
4	Check the complete compressor support brackets for their healthiness.	check ok		
5	Visually examine the compressor mounting resilient blocks and take needful action.	check ok		
6	Visually examine the various interconnecting pipelines and its support clamps of the compressors for their healthiness / airleakage.	check ok		
7	Cleaning of input air filter and the primary oil filter.			
8	Clean the breather assembly	check -		
9	Check and Drain the inter cooler / after cooler assembly and also observe its discharge and take needful action.	check		
10	Ensure healthiness / intactness of various protection guards of compressor.	check		
11	Check the condition of the coupling between compressor & motor	check		
12	Visually check the compressor fan blade for their healthiness	check		
13	Check the oil level in the compressor and top up as per requirement.	check		
14	Carry out the inspection of each type of compressor in line with recommendation of respective manufacturer.	check		
15	Check compressor foundation base for crack (For under slung compressor)	check		
B	Aux. Compressor (CPA)			
1	Clean the CPA externally. Clean Suction Air filter.	check		
2	Check oil level and condition of oil and top up if required with the oil specified (Servo Press 150).	check		
3	Check tightness of all nut bolts, foundation bolts.	check		
4	Measure and record time & pressure of the auxiliary compressor	check		
C	Sch IC: i) CP all discharge valves to remove.	check		
	ii) CP all discharge valves refit by overhauled	check		
	iii) CP oil to be change	check		
	iv) Baby compressor/CPA oil to be change	निकाला जा रहा है		
	CPA			
	CP-1	Changed		
	CP-2	Changer		
	v) Baby compressor/CPA safety valve (8.5 kg/cm ²) to be replace by overhauled	check		
D	AIR SUPPLY AND PNEUMATIC SYSTEM			
1	NRV and unloader valves fail due to ingress of Carbon which extracted from CP. So, one cycle dismantling, checking and cleaning of NRV and unloader to be ensured in all type of 3-phase locomotives.	check		
E	AIR DRYER			
1	Inspect the air dryer for any damage and its mounting for any damage.	check ok		

क्र.सं. SN	कार्यवाही कार्य / विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
	crack in mounting plate, intactness of fasteners. Check for any hit mark on main reservoir and drain cock. Check for proper position of the air dryer angle cocks.	chalo		
2	Check the integrity of the welding of the hanging support.	chalo		
3	Check humidity indicators for blue color. White indications mean the air dryer is not operating effectively. Desiccant shall be changed, if needed.	chalo		mandu 23.9.24
4	Replace desiccant if found to be contaminated with either water or oil.	chalo		
5	Testing of air dryer for proper functioning to be done as per Technical Circular no. RDSO/2011/EL/TC/0108, Rev.'0' dtd. 7.3.11 for testing of air dryers and its efficacy on electric locomotives.			
F	E-70 BRAKE SYSTEM (PNEUMATIC PANEL)			
1	Visually inspect the pneumatic panel noting any sign of air leakages and rectify accordingly.			
2	Clean the pneumatic panel externally.			
3	Check for correct position of the cut out/isolating cock.	chalo	mandu	
4	Check all the cable connections for looseness or deficiency of cleats. If any inspection section is to be informed (IC Sch)			
5	Ensure provisioning and proper functioning of RS gauge. Switch, lamp & other gauges on pneumatic panel.			
6	Check the sealing of connectors. Flexible pipe, regulator, air tank etc.			
G	BRAKE ISOLATING COCKS: Check for correct operations of BP & FP angle cocks. Check the condition of BP/FP pipe, cock & isolating handle for any hit mark or crack. Check correct condition of Bogie Isolating cock.	chalo	mandu	
H	BRAKE HOSES: Examine the brake hoses at the ends of the locomotive to make sure that they are not chafed or damaged. Check coupling cocks for correct operation.	chalo	23.9.24	
I	AIR RESERVOIR			
1	Blow down and drain the reservoir until exhaust of complete water and oil. Check all drain cocks are closed on completion.	chalo		
2	Check for any hit mark on main reservoir & drain cock.	chalo		
3	Ensure intactness / tightness of main reservoir foundation bolts.	chalo		
4	Check the rubber gasket at the inspection cover of air reservoir for any damage/leakage.	chalo		
5	Check the pressure regulator operates correctly.			
J	THROTTLE VALVE (IC Sch): Overhaul the throttle valve and change the rubber parts.			
K	IN LINE FILTER (IC Sch): Remove the filter element and clean thoroughly.			
L	HORN & HORN SWITCH			
1	Check condition of horn handle for any damage. Check for proper operation of horn.	chalo		
2	Check the condition of horn boots and replace if required.			
3	Remove the horn & horn switch and Overhaul. Replace the diaphragm of horn. (IC Sch)			
4	Check all angle cocks for proper operation.	chalo		
5	Ensure proper colour code of all the angle cocks.	chalo		
6	Check the tightness of the brake electronics control cards holding screws.			
7	Check all the pipes for leakage or damage. Check all the clamps for any damage and looseness.			
M	वायुवीय प्रणाली PNEUMATIC SYSTEM			
1	MR Safety valve, ADC, Panto throttle valve (Only for conventional Pantograph) replace by Overhaul in IC Sch.			

Signature and Name of Technician

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
CHECK SHEET FOR 3PHASE HR PANTOGRAPH ASSEMBLY (IA/IB/IC)

लोको संख्या Loco No 32472 क्लास Class IB शिड्यूल में लेने का दिनांक Date taken under Sch 98

SCHEDULE: PANTOGRAPH SR.NO. /Make//Type: LOCATION:

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
1.	Ensure that wearing strips are without sharp edges and no gap between end strip and wearing strip.	Smooth and no gap	all	all	<u>Mamoy Shetty</u> <u>25-9-21</u>
2.	Ensure availability of bolts, nuts, lock nuts, spring, washer, split pin etc	All intact	all	all	
3.	Check the rubber air pipe for any leakage	No leakage	all	all	
4.	Check the rubber air bellow for any leakage/damage/rubbing	No leakage	all	all	
5.	Check lower frame assembly for any crack and play	No crack	all	all	
6.	Check upper frame assembly for any crack and breakage	No crack	all	all	
7.	Check coupling rod assembly for crack and its fitting intactness.	No crack & intact	all	all	
8.	Check diagonal rod for any crack both end side. (by DPT)	No crack	all	all	
9.	Check plain bearing eye bush for freeness	free	all	all	
10.	Check ball bearing friction of Base frame, Upper frame and Coupling rod.	ok	all	all	
11.	Check Cam wear on top of cable rope guide surface.	OK	all	all	
12.	Check condition of all shunts (not more than 5% strand broken)	OK	all	all	
13.	Check pantograph pan for any external hitting marks	No hitting marks	all	all	
14.	Verify all welded joint for cracks.	OK	all	all	
15.	Check the control box for any air leakage	No air leakage	all	all	
16.	Check the condition and operation of hydraulic damper.	Normal	all	all	
17.	Check Parallel guide bar for free function of Panto Pan Head	working	all	all	
18.	Ensure that the thickness of wearing strips above the condemning size	New - 39 mm Condemn - 26 mm	37-52mm 38.10mm	38-mm 37.40mm	
19.	Check the static contact force of pantograph at 6.4 kg/cm ² MR pressure	70 N / 7Kg Load Test	all	all	
20.	Check the raising and lowering time of pantograph	6 to 15 second	all	all	
21.	Check the horizontality of panto pan with spirit level.	Level OK	all	all	
22.	Check the function of auto drop device (ADD)	Working	all	all	

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
23.	Check the function of over reach detector (ORD)	Working	Done	Done	Money Shores 25-9-24
24.	Check air filter for air leakage or not draining.	No leakage	all	all	
25.	Check the all pneumatic valve (3/2 valve, throttle valve and safety valve) for any leakage and proper function.	No leakage	all	all	
26.	Check pressure regulator for any leakage and proper function.	No leakage	all	all	
27.	Check the tightness of all bolts and thread locking compound to be provided in bearing base bolts.	Provided	all	all	
28.	Examine the roofline conductors for defects/looseness at joints.	No defect	all	all	
29.	Check the roof for any foreign material such as wire / clothpieces.	No foreign material	all	all	
30.	Check tightness, cleanliness and wear-tear of shunt connection between Panto and Roof line.	Tight, clean	all	all	
31.	Check that the pan head locking pins springs are in place and secure. Check that the head moves freely on its mounting pins attached to the apex frame. If movement is not free, check that the head is not twisted, the side beam are not bent/twisted & mounting pins are not loose or bent.	OK	all	all	
32.	Check tightness of the flexible shunt connecting screws. When a loose screw is found the shunt should be disconnected from the head &/or the pantograph frame, the mating surface cleaned & the shunt reconnected with new screws. Check all shunts for frayed strands and renew any shunt in which 5% or more of the strands are frayed.	OK	all	all	
33.	IC Sch: Clean filter element, if found brownish then replace with new.	Clear	clear	clear	

Remarks if any:

Signature and Name of Technician

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC Schedule of 3 Phase Electric Locomotive [Heavy Section]

लोको संख्या Loco No 32456 क्लास Class 2B शिड्यूल में लेनेका दिनांक Date taken under Sch 6/8

Sr. No	Details of Work / Inspection (Schedule)	Action Taken	Name of TCN	Sign / Remarks
A	VCB			
1	Check insulator for crack, VCB roof bus bar, chips and flash marks. If damage in service, renew it	Check		
2	Clean insulator, other parts that are dirty with soft, clean & dry cloth. (Do not use cleaning products based on fluoridated or chlorate compounds or sodium meta silicate), apply Silicon Spray.	Check		
3	Check contact spring (Earthing contact)	Check		
4	Check moving contact lever of auxiliary switch and tension spring & its mounting bolts for breakage	Check		
5	Check the cables & lugs for broken cables, lugs looseness	Check		
6	Check the main bushing condition, its connection and foundation bolts and clamps.	Check		
7	Check the tightness of HV Connections (Torque 70Nm)	Check		
	IC Sch:			
	a) Check for damage to connection of the earthing isolator, Cleaning & greasing, if necessary.	Check		
8	b) Check the torque of VCB fixing screw (Torque 70Nm)	Check		
	c) Remove the bottom cover and check that the control air and electrical connection are tightened & Ensure that there is no leakage from pressure regulator	Check		
B	MAIN TRANSFORMER			
1	Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage. Clean the gauge with dry cloth	Check		
2	Inspect the colour of the silica gel. If it is pink, replace with blue silica gel	Check		
3	Examine the flanges of the pipe couplings and flexible hose that link the transformer and conservator	Check		
4	Check the availability of hosepipe over the oil pipe compensator and check stoochi coupling pipes for proper layout/fitment.	Check		
5	Check the main Transformer and its protection cover for any damage, crack & oil leakage. Also refer instructions contained in RDSO's letter no. EL/3.2.1/3-Ph / TC-0076 for arresting oil leakages cases"	Check		
6	Check the deformity of Transformer drain cock protection cover guard if deformed replace it	Check		
7	Perform the sample test on transformer oil. Check specific value of BDV, DGA, moisture and acidity and condition monitoring of loco transformer by Dissolve Gas Analysis. (RDSO/SMI/138 & OEM Doc. Dt. 27th Nov, 1995)	Check		
8	Check visually the condition of transformer foundation bolt and Nylon lock nuts for proper locking.	Check		
9	Check the Ecable. Clean and visually inspect the insulator condition and Electrical connection of transformer and converter for crack & flashed mark and ensure red marking	Check		
10	Visually inspect for leakage, damage, burning/overheat signs and all fixing clamps condition of oil cooling metallic pipes, Prismatic level gauge, TFP bushing, HV bushing, Hose pipe etc.	Check		

Mangesh Kumar
25-9-24

24/9/24

Signature and Name of Technician

Signature and Name of JE/SSE

Sr. No	Details of Work / Inspection (Schedule)				Action Taken	Name of TCN	Sign / Remarks
C	EARTHING CONNECTION.						
	a) Check the earthing connection tightening torque 70Nm.						
	b) Connection of earthing system in 3-phase locomotives type WAP5, WAP7 & WAG9 (As per RDSO/2007/EL/SIM/248 Rev '0' dated 22.11.2007)						
	c) Check Earthing shunt of Bogie to Body						
	Axle box 1-3	✓	Axle box 2-4	✓			
	Axle box 5-7	✓	Axle box 6-8	✓			
	d) Check Earthing shunt of Axle to Bogie						
	Axle box 2	✓	Axle box 3	✓			
	Axle box 6	✓	Axle box 7	✓			

Panto Refrio during free sch.

① Panto & Pneumatic Box (check by panto section Because Panto lowering time excess

② CP-A Dummy Detec line (Roundry) After Bregeing to fild

MPL TIME 0 TO 10 KG / cm^2 AFTER CP Repair attended.
3 MIN 08 SEC.

Signature and Name of Technician

Signature and Name of JE/SSE